TEST PLAN FOR LMS
DSA PROJECT

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NIM-BSEE-2021-19

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1.TEST LOGIN FUNCTIONALITY:

Test logging in as a librarian/student valid credentials.

```
*******************************************************************************
              Welcome to Namal Library Management System
*******************************************************************************
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3): 1
librarian Login
Enter your username: namal
Enter your password: 123
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6):
```

ii. Test logging in as a librarian/Student with in-valid credentials:

```
*******************************************************************************
              Welcome to Namal Library Management System
*******************************************************************************
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3): 1
librarian Login
Enter your username: namal
Enter your password: 34drf
Invalid credentials. Access denied.
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3):
```

2.TEST ADD BOOK FUNCTIONALITY:

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 1
Enter the book name: Pakistan
Enter the book author: Imran Khan Niazi
Enter the book subject: A personal history
Book added successfully!
```

- Verify that the book is correctly added to the catalog and saved to the file.

```
Title: CP
author: Sir Bacha
subject: Computer Programming
**********

Title: PHY
author: Sir Zia
subject: Physics
**********

Title: QS-2
author: Sir Yaseen
subject: Quranic Studies
**********

Title: The last hope
author: Ahti.M
subject: A personal history
**********

Title: Pakistan
author: Imran Khan Niazi
subject: A personal history
**********
```

3. TEST REMOVE BOOK FUNCTIONALITY:

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 2
Enter the book name: The last hope
Book removed successfully!
Title: The last hope
 Author: Ahti.M
 Subject: A personal history
```

Verify that the book is correctly removed from the catalog and saved to the file.

```
Title: ENAA
author: Sir Usama
subject: Electric Network
**********

Title: CP
author: Sir Bacha
subject: Computer Programming
**********

Title: PHY
author: Sir Zia
subject: Physics
**********

Title: QS-2
author: Sir Yaseen
subject: Quranic Studies
**********

Title: Pakistan
author: Imran Khan Niazi
subject: A personal history
**********
```

4. TEST MODIFY BOOK FUNCTIONALITY:

- Test modifying the details of a book in the library catalog.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 4
Enter the book name to be modified: CP
Enter the new book name: Computer Architecture
Book modified successfully!
```

- Verify that the book details are correctly updated and saved to the file.

```
Title: ENAA
author: Sir Usama
subject: Electric Network
**********

Title: Computer Architecture
author: Sir Bacha
subject: Computer Programming
**********

Title: PHY
author: Sir Zia
subject: Physics
**********

Title: QS-2
author: Sir Yaseen
subject: Quranic Studies
**********

Title: Pakistan
author: Imran Khan Niazi
subject: A personal history
**********
```

5. TEST SEARCH BOOK FUNCTIONALITY:

- Test searching for books by title, author, and subject.
-

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 3
Enter the book name: Pakistan
Book found in the library.
Title: Pakistan
author: Imran Khan Niazi
subject: A personal history
```

6. TEST WRONG INPUT FUNCTIONALITY:

In case the user input the wrong input then the program will ask the user to right the correct input.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 9hk3
Invalid choice! Please enter a number from 1 to 6.
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6):
```

7.TEST LOGOUT FUNCTIONALITY:

If the user wants to logout as librarian and want to access has student or vice versa then the user has to enter 6. The program will take out of the current profile.

```
Library Management System
1:- Add Book
2:- Remove Book
3:- Search Book
4:- Modify Book
5:- Display Books
6:- Exit
Enter your choice (1-6): 6
Logging out as Librarian
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3):
```

8. TEST CLOSE PROGRAM FUNCTIONALITY:

-To close the LMS program, the user has to click on 3 and a final message will be displayed and user will be come out of the program.

```
Login as:
1:- Librarain
2:- Student
3:- Close Program
Enter your choice (1 to 3): 3
Exiting the program...
Thank You for visiting Namal Library!
```